PNEUMATIC TESTING PROCEDURE (N2)

(Doc No. CT8-PRO-PI-007)

USA PROJECT UNIT 1 - TALLGRASS

Revision History				
Revisions	*Date*	*Descriptions*	*Prepared By*	*Approved By*
0	25-Mar-26	First Issue	Ms. Michelle Wilrutstien	Mr. Phisit Pew-la-or

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1. PURPOSE AND SCOPE

This procedure provides the guideline and precautions to be taken for Pressure Testing, Flushing and Drying of the Piping for the Modularization works, Convent. This procedure is applicable for both Piping installed on modules and also the loose spools.

2. DEFINITION

OWNER	: Mitsubishi Power Americas Limited (MPW-A)
CLIENT	: Cimtas (Ningbo) Steel Processing Co., Ltd. (CIMTAS)
FABRICATOR	: BJC Heavy Industries Public Company Limited (BJCHI)
ITP	: Inspection and Test Plan

3. REFERENCE

ASME B31.1	: Power Piping

4. RESPONSIBLITY

Project Manager has the overall authority and responsibility for all activities required to ensure that the Construction work complies with Project Requirements.

Production Manager shall be responsible for ensuring that testing activities for the piping shall always be performed by trained, experienced employees in a manner acceptable to project specification and procedure requirements.

HS&E Manager is responsible for ensuring compliance of all parties with the HSE requirements of the project. He will provide trained and competent personnel to monitor and assist with compliance by construction personnel with the HSE requirements and will assist with development JSA.

QA/QC Manager/ Engineer are responsible for ensuring the all-construction inspection and testing requirements has been issued prior to commencement for testing activities. All quality assurance and control activities are carried out in accordance with project requirements.

QA/QC inspector are responsible for ensuring that no impeditive items on Punch List is present prior to Pneumatic test. QA/QC inspector are responsible for ensure that pressure test pumps and accessories are inspected prior to works (Check List required). All pressure test devices, hoses, accessories shall have a pressure rating equal to or higher than the line to be tested.

Contractor shall be responsible to manage and perform all activity to complete the test, contractor shall nominate a qualified testing engineer. He shall be responsible for all testing and associated work.

Client and if required Owner shall witness the Pressure Testing and sign off the Pressure testing reports.

5. EXECUTION

5.1 SAFETY AND ENVIRONMENTAL CONSIDERATIONS

- Entry into the testing area shall be restricted to prevent unauthorized access during testing Suitable means of restriction access include, but not limited to, roping or taping off the area, or erection barriers.
- No work shall be permitted on sections under test or the associated connections until the pressure is reduced to a safe level and released by operator.
- Danger signs shall be also be placed and visible at each entry point.
- Personnel shall protect themselves with suitable shields and apparel such as hard hats, goggles, hearing protection, and other required safety items.
- Verify the location of all blanks, the required material and thickness specified in the fabrication, construction drawings, pressure test drawings. Verify proper test manifold design including review of all materials and component ratings against available test fluid.
- Pressure testing shall be performed in accordance with the safe working practices mentioned in HSE manual of the project.
- Permit to work (PTW) shall be obtained prior to start of any activity.
- Information shall be given to all disciplines.
- Safety precautions shall meet the requirements of all applicable safety rules and regulations. Since the pressure during the pressure test is higher than normal operating conditions, the following safety precautions shall be observed particularly.
- Area shall be barricaded with warning tapes and warning signs shall be placed.
- Approved site notice shall be displayed prior to pressure test.
- Entrapped air shall be vented out from the pipe to minimize the risk of high energy release.
- Trained and competent personnel only shall be deployed for test.
- Only the test crew shall be allowed to enter within 5m of pipe line under pressure.
- Visual inspection for leakage shall not be carried out when the pressure is being raised or within 15 minutes thereafter.
- No activity shall be permitted on the test lines under pressure.
- Any manifold fabricated for pressure test purposes shall be tested separately from any other piping before it is used in conjunction with testing of any system.
- The manifold shall be tested to a pressure not less than twenty percent above the test pressure to be applied on the system to be tested.
- Relevant certificates for instrument, equipment and for the competency of personnel shall be submitted together with application for PTW.
- Test/calibration certificates of pressure gauges shall be submitted.

5.2 TEST AND MEASURING EQUIPMENT

All components used for the Pressure testing such as manifold, Hoses, Gauges, Chart recorders, etc. are calibrated by independent laboratory. Test blinds shall be of suitable material with traceability to Test certificates. The thickness of blinds shall be not less than tm.

Gage Range. When dial indication and recording pressure gage(s) are used for testing, they should preferably have the dial(s) graduated over range of approximately double the intended maximum pressure, the full range of the pressure gauge shall be between 1.5 and 4 times of test pressure

Gages shall be calibrated against a standard deadweight tester, a calibrated master gage, or a mercury column.

1) Calibration shall not exceed 12 months.
2) Gages diameter minimum of 75 mm.
3) Gages resolution shall be minimum reading of about 5% or more of test pressure.
4) Mininum of 2 gages shall be used. One at the hightest position possible and other at the lowest system position

5.3 INSPECTION / TEST PREPARATION

Before preceding the test, contractor shall prepare test package which will include mentioned documents in the next page.

Whenever possible the Package should be performed by pipeline System or Sub-systems preferably instead of individual pipes. (When the System or Sub-system extends beyond construction module and there is no clear battery limit (flanges/ threaded connection/welded connection/shut-off valve/etc.) the scope of the test should be limited by the closest blocking device/accessory)

The pressure test shall be performed after the removal of dirt and construction debris. The Package shall be submitted to COMPANY for approval.

- Test Pack Cover sheet
- Test Package Content Check List
- Marked up P&ID
- Marked up Isometrics List and Pipe Support Drawings
- Welding History Sheet
- NDE reports
- Flange Joint Assembly and spiral wound gasket reports (alignment, Orifice flange report, if applicable)
- Dimension verification records

- Pre-pressure Test Inspection check report
- Calibration certificates of pressure gauges
- Pressure Test report
- Painting Inspection Reports (Touch-up)
- Insulation Checklist
- Pipe cleanliness and Verification Records
- Piping Restoration Checklist
- Punch List

Equipment or any piping system that not to be included in flushing and testing shall be either removed or isolated. All items that subject to temporary isolation/removal shall be positively recorded on a piping isolation record before flushing and testing.

The piping system subject to pressure testing shall be disconnected from other piping system or isolated at each different test pressure.

Spool pieces, if removed from the system for pressure testing, shall be tested separately.

Joints Exposed, all joints (including structural attachment welds to pressure-containing components), and bonds shall be left uninsulated and exposed for examination during leak testing, except that joints previously tested may be insulated or covered.

Piping be provided with additional temporary supports, if necessary.

5.4 PROCESS

5.4.1 Precaution

- Calibration certificates shall be made available to the inspection personnel prior to commencement of test.
- Calibration certificate of pressure gauges shall be made available to COMPANY
- Spare pressure gauges shall be prepared to avoid any delay due to calibration.
- Isolation valves shall be provided at the point where the test manifold concerns to the piping under pressure test.

5.4.2 Preparation of piping system for testing

- Pressure test coordinator shall follow up the sign off of As-Built drawings, weld inspection summary and “A” point in the punch list.
- It shall be checked that gauges used for testing are calibrated and of required shall be 1.5 to 4 times the pressure range.
- No equipment such as expansion joints. Filters etc., designed for test pressure and shall be isolated from system under test.

- On completion of mechanical works for the test circuit, it shall be isolated from other lines by means of blind flanges or spades.
- Piping and equipment to the test system being tested shall be protected from accidental pressurizing by opening the equipment to atmosphere.
- Necessary precautions shall be taken to ensure that no damage is caused to in-line instrumentation. In-line instrumentation shall be removed prior to test and pressure check list to be field.
- A minimum of two pressure gauges shall be used position at highest and lowest point at circuit.
- Thickness of test blinds shall be not less than **" *t* "**, the minimum of blind shall be calculation as per following formulation:

Basic Formula

The thickness t of a blind flange can be estimated using the following formula, which is a modified version of the general formula for pressure vessel design:

$$t = \frac{P \cdot D}{2 \cdot S \cdot E - 0.6 \cdot P}$$

where:

- P = Design pressure (in psi or MPa)
- D = Diameter of the flange (in inches or mm)
- S = Allowable stress of the material (in psi or MPa)
- E = Joint efficiency (typically 1 for new or well-welded flanges)

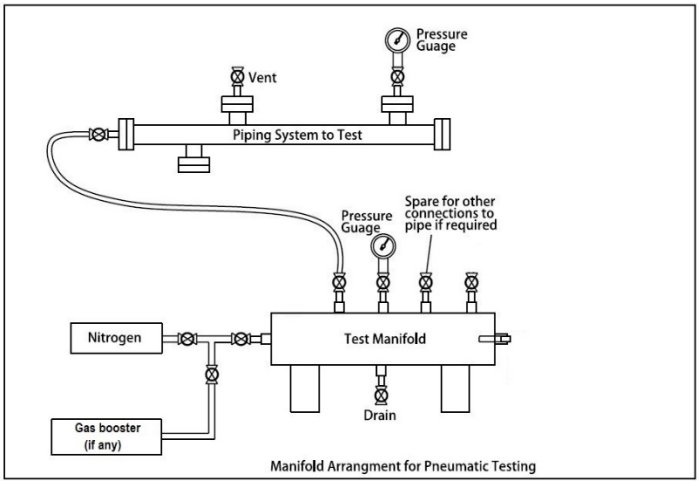
- All test blinds shall be provided with a lug or handle extending at least 100 mm. and not more than 200 mm. beyond the flange perimeter which shall be painted bright red.
  - Temporary blinds shall be adequately provided with gaskets to prevent damage to flange facings.
  - Lines containing check valves shall have the source of pressure located in the piping upstream of the check valve so that the pressure is applied under the seat. If this is not possible, their internals shall be securely blocked open or removed.

5.4.3 Pneumatic test

- The test medium shall be Nitrogen (N2).
- *Test pressure shall be 1.2 times the maximum working pressure.*
- The pressure in the system shall gradually be increased to not more than one-half of the test pressure, after which the pressure shall be increased in steps of approximately one-tenth of the test pressure until the required test pressure has been reached at approximately 50% and 80% of Test Pressure, preliminary check for leakage shall be performed. The pressure shall be continuously maintained for a minimum time of 10 min. It shall then be reduced to the lesser of design pressure or 100 psig [700 kPa (gage)] and held for such time as may be necessary to conduct the examination for leakage. Examination for leakage detected by soap bubble or equivalent method shall be made of all joints and connections. The piping system, shall show no evidence of leaking.
- No access shall be allowed for anybody in the area when the test system is pressurized.
- Access to the test area shall only be allowed only after reducing the test pressure to design pressure at this pressure piping defamations and cracks shall be checked. Pressure test report shall be filled.
- Pneumatic test for reinforcing pads shall be done in accordance with the project.

5.4.4 Reporting

The pressure test shall be recorded and reported using form F-QA-173 Rev.1 in appendix 2. Photos record shall be attached to the report.



Manifold Arrangment for Pneumatic Testing

APPENDIX 1: Safety areas for pneumatic testing

Safety areas for pneumatic testing

For pressure vessels the safety distance shall be according Table B.1.

For piping the safety distance shall be in accordance with Table B.1 and upon consultation with the parties involved.

Table B.1: Safety distances for vessels

	Safety Distance [m]							
Volume [m^3]	P_t=5 [bar]	P_t=10 [bar]	P_t=15 [bar]	P_t=20 [bar]	P_t=25 [bar]	P_t=30 [bar]	P_t=35 [bar]	P_t=40 [bar]
1	2.1	3,3	4.3	5.1	5.9	6.5	7.1	7.7
2	2.9	4.7	6.1	7.2	8.3	9.2	10.1	10.9
3	3.6	5.7	7.4	8.9	10.1	11.3	12.4	13.4
4	4.1	6.6	8.6	10.2	11.7	13.0	14.3	15.4
5	4.6	7.4	9.6	11.5	13.1	14.6	16.0	17.2
7	5.4	8.8	11.4	13.6	15.5	17.3	18.9	20.4
10	6.5	10.5	13.6	16.2	18.5	20.6	22.6	24.4
15	7.9	12.8	16.6	19.8	22.7	25.3	27.6	29.9
20	9.2	14.8	19.2	22.9	26.2	29.2	31.9	34.5
30	11.2	18.2	23.5	28.1	32.1	35.7	39.1	42.2
40	13.0	21.0	27.2	32.4	37.0	41.3	45.1	48.8
50	14.5	23.5	30.4	36.2	41.4	46.1	50.5	54.5
60	15.9	25.7	33.3	39.7	45.4	50.5	55.3	59.7
70	17.2	27.7	35.9	42.9	49.0	54.6	59.7	64.5
80	18.3	29.7	38.4	45.8	52.4	58.3	63.8	69.0
90	19.5	31.5	40.7	48.6	55.6	61.9	67.7	73.1
100	20.5	33.2	42.9	51.2	58.6	65.2	71.4	77.1
150	25.1	40.6	52.6	62.7	71.7	79.9	87.4	94.4
200	29.0	46.9	60.7	72.4	82.8	92.2	100.9	109.0
300	35.5	57.4	74.4	88.7	101.4	113.0	123.6	133.5
400	41.0	66.3	85.9	102.5	117.1	130.5	142.7	154.2
500	45.9	74.2	96.0	114.5	131.0	145.8	159.6	172.4

Note:
The values shown in the table are based on the following formula acc. to Regels voor toestellen onder druk "Rules for pressure vessels" T 0240:

$$l = 1.51\sqrt{V\left(p_t - p_t^{0.714}\right)}$$

in which: l = safety distance
V = volume
p_t = test pressure

Intermediate values can be interpolated.

APPENDIX 2: Pressure Test Report Format

BJCHI	PRESSURE TEST REPORT	cimtas | ningbo

Client :			Report No. :	
Contractor :			Test Date :	
Project Name :			Location :	
Project No. :			Test Medium :	
Description :			Chloride Content :	
Procedure No. :			Graph Record No. :	
Test Package No. :			Module No. :	
Drawing No. :			Unit No. :	
Spool No. :				
System/Service :				
Pressure Gauge No.	1st Gauge :		Digital Recorder No. :	
	2nd Gauge :		Certificate No. :	
	3rd Gauge :		Calibration Date :	
Certificate No.	1st Gauge :		Design Pressure :	KpaG
	2nd Gauge :		Test Pressure :	KpaG
	3rd Gauge :		Test Duration :	Minutes
Calibration Date	1st Gauge :			
	2nd Gauge :			
	3rd Gauge :			
Duration of Test	Start Time :		Actual Test Pressure Recorde	KpaG
	Finish Time:		Actual Test Pressure Pressure Gauge	Bar
Holding Time		Minutes	Test Results :	☐ Satisfactory
Test Temperature				☐ Unsatisfactory

REVIEWED & WITNESSED			
COMPLETED BY	BJCHI	CIMTAS	OWNER REPRESENTATIVE
NAME :			
POSITION TITLE :			
SIGNATURE :			
DATE :			

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